

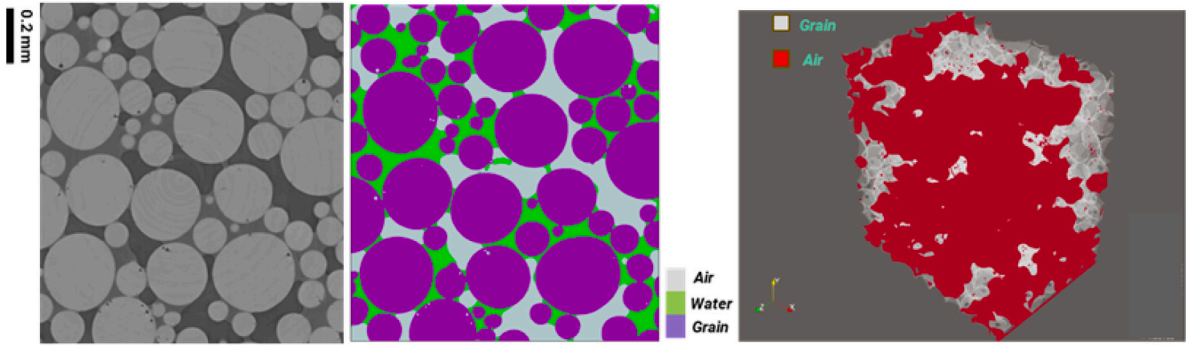


Fig. 1. (a) micro-CT (greyscale) cross-section of dry glass bead, (b) 2D segmented image, (c) 3D visualization of the air phase during primary drainage at $S_w = 0.537$, $A_{nw} = 30.63 \text{ mm}^2$ and $P_c = 10.3 \text{ Pa}$.

provided in Supporting Information 1 (SI-1). The X-ray projections from each scan were reconstructed into 3D images, allowing detailed visualization of the displacement processes. Subsequent image processing was conducted using AVIZO software, where a non-local means filter was applied to de-noise the images (Buades et al., 2005; Tristán-Vega et al., 2012). The images were segmented into grains, brine, and air using a watershed-based segmentation technique (Beucher, 1979) (Fig. 1). The segmented images were then used to measure S_w , A_{nw} , and χ_n . An uncertainty analysis was performed by varying greyscale thresholds for phase segmentation. The maximum deviation in brine saturation (S_w) was 2%, indicating a minimal impact on the interpretation of the segmentation based on the watershed method. An example of the thresholding procedure in AVIZO is provided in Supporting Information 1 (SI-1).

S_w and A_{nw} were computed using AVIZO software, while χ_n of the nonwetting phase was calculated using the following equation:

$$\chi = N - L + O, \quad (1)$$

where N is the number of isolated clusters in a fluid phase, L is the number of redundant connections within all clusters, and O is the number of cavities, e.g., isolated background clusters completely enclosed by the phase of interest (Schlüter et al., 2016).

Next, the resulting Euler characteristic of the nonwetting phase (χ_n) was normalized and fitted using the power-law scaling proposed by Schlüter et al. (2016) (Eq. (2)). The normalized Euler characteristic of the nonwetting phase, $\hat{X}$, has been introduced to quantify the effect of fluid topology on nonwetting phase (NWP) trapping (Herring et al., 2013, 2015). Normalization is achieved by dividing χ_n at a given nonwetting phase saturation (S_n) by the Euler characteristic of the fully saturated pore space, so that $\hat{X}$ reaches unity at full NWP saturation. Schlüter et al. (2016) demonstrated that $\hat{X}$ follows a power-law relationship with S_n (Eq. (2)):

$$\hat{X} = pS_n^k + \epsilon, \quad (2)$$

where k is the characteristic of the slope of $\hat{X}$ as a function of nonwetting phase (NWP) saturation, S_n , and p , ϵ are the additional, free parameters. The exponent k turns out to be characteristic for the flow regime and independent of the underlying porous medium.

2.3. Oil–Brine System Data: Schlüter et al. (2016)

Our experimental methodology closely follows the framework established by Schlüter et al. (2016), who used glass beads as a porous medium in multiphase flow experiments. Therefore, we leveraged their oil-brine system data, which are comparable to our gas-brine conditions and porous medium, to strengthen the relevance and interpretation of our results. The experiential work of Schlüter et al. (2016) relies on micro-CT imaging to investigate pore-scale interactions and compute geometric parameters. Schlüter et al. (2016) used glass beads with a

size distribution of 35% 0.6 mm, 35% 0.8 mm, and 30% 1.0–1.4 mm diameter, with a porosity of 0.33. The characteristics of these glass beads are comparable to those of our glass bead sample mentioned in Section 2.1. For the case of interfacial tension (IFT), Schlüter et al. (2016) reported a value of 26 mN/m for their dodecane-brine system, whereas the IFT of the air-brine system in our study is 72 mN/m (Pallas and Pethica, 1983). The similarity in glass bead samples is crucial for validating our observations and extending the applicability of the findings by Schlüter et al. (2016) from oil-brine to gas-brine systems.

2.4. Relative permeability calculations

We used the pore-scale finite-volume solver (PFVS) to compute the effective permeability (k_e) of each segmented fluid phase (Chung et al., 2019). This image-based approach to relative permeability was first proposed and validated by Berg et al. (2016) and has since been widely adopted in pore-scale studies (Gao et al., 2019; Jetttestuen et al., 2024; Lin et al., 2019b,a; Liu et al., 2017; Zou et al., 2018; Zoeir et al., 2019).

For each dataset, binary images were generated in which the target phase (e.g., oil in an oil-brine system) was assigned as the conducting medium, while all other phases were treated as solid. The PFVS then determined the steady-state pressure distribution by solving an elliptic equation for incompressible Stokes flow, with local conductivities derived directly from voxel geometry and connectivity. The resulting effective permeability of the target phase was normalized by the absolute permeability (k) of the same image volume to obtain the relative permeability (k_r).

Because the segmented phase geometries were obtained directly from micro-CT images acquired after controlled drainage and imbibition steps, the effects of pore geometry, wettability, and interfacial tension were inherently incorporated into the calculated permeabilities.

3. Results

This section presents a comparative analysis of multiphase flow in oil-brine and gas-brine glass bead packs, examining both continuum-scale and pore-scale behaviors. We begin by presenting representative pore-scale images for each displacement process to visually illustrate the observed hysteresis behavior. We then compared the k_r curves of the oil-brine system, calculated based on the images available from Schlüter et al. (2016), and the gas-brine system based on our gas-brine experimental images. Next, we analyze the capillary pressure relationship and key fluid morphological parameters, including A_{nw} and χ_n as functions of S_w . These insights reveal how fluid properties and pore-scale displacement mechanisms drive differences in multiphase flow behavior, providing a deeper understanding of the observed hysteresis behaviors.

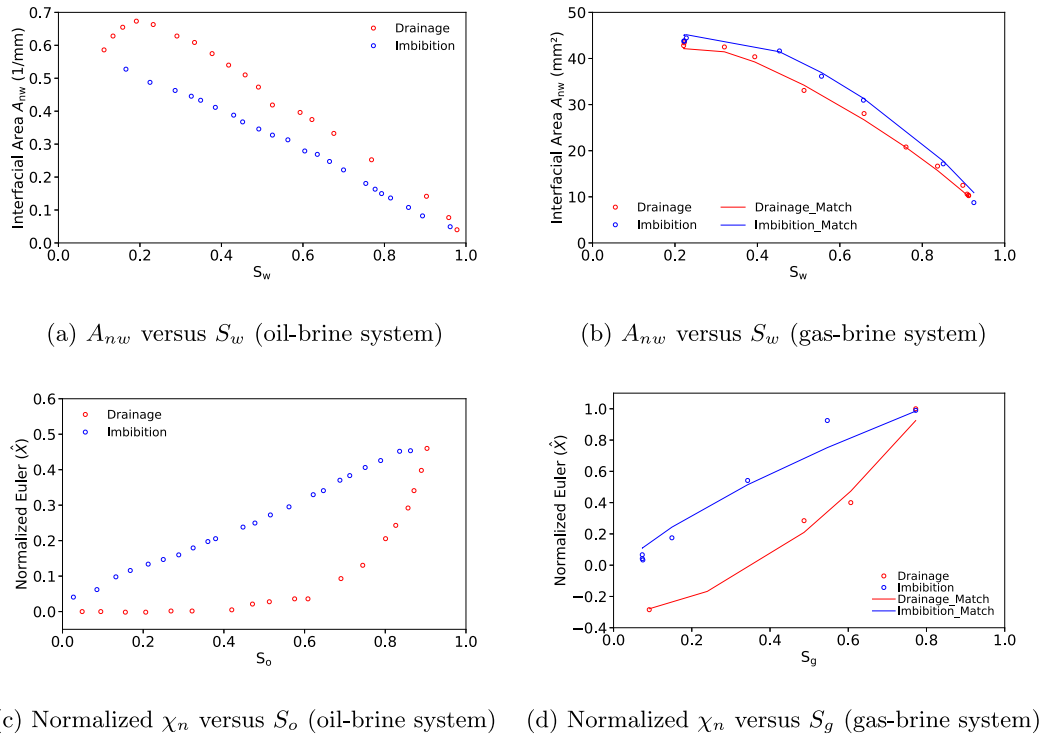


Fig. 5. Comparison of fluid morphologies of the oil-brine and gas-brine systems.

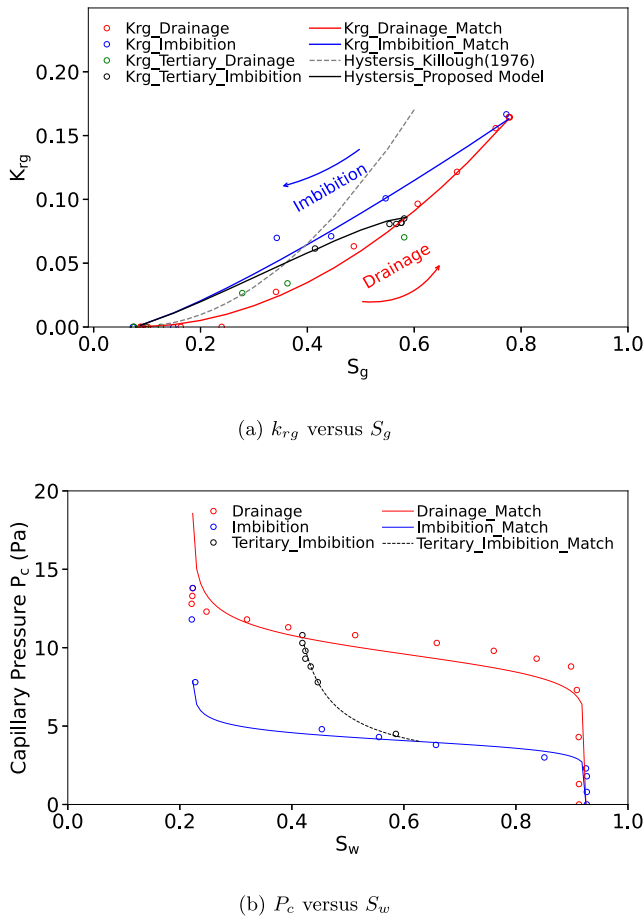


Fig. 6. (a) Proposed model for gas-brine k_r hysteresis, (b) capillary pressure hysteresis.

4. Discussions

The experimental findings of this work reveal notable differences in the behavior of gas-brine and oil-brine systems within glass-bead porous media. Although this reversal behavior in k_r trends has been observed in prior studies, it has not been extensively analyzed from a pore-scale perspective, which is addressed in this work.

Early investigations by Naar et al. (1962) first identified this phenomenon in unconsolidated porous media, where k_r during imbibition exceeded that observed in drainage. Their experiments demonstrated that the flow behaviors of consolidated and unconsolidated porous media differ significantly. Specifically, in consolidated rocks, the k_r of the nonwetting phase during drainage was higher than that during imbibition. In contrast, for unconsolidated media, the opposite trend was observed, where the imbibition of the nonwetting k_r was greater than that in drainage. This reversal behavior was attributed to differences in the distribution of pore size and the extent of cementation within the medium (Naar et al., 1962). Furthermore, in addition to the pore structure, our work demonstrates that the phases used are important. For unconsolidated glass beads, the reverse behavior was observed only for the gas-liquid experiment.

Jerauld and Salter (1990) reported similar trends in unconsolidated materials, where the results indicated that the nonwetting phase exhibited higher k_r during imbibition than in primary drainage. They linked this behavior primarily to the microscopic arrangement of fluids within the pore space, emphasizing that the pore-body to pore-throat aspect ratio plays a crucial role in the hysteresis patterns. Specifically, they found that the variations in consolidation could be understood in terms of changes in the aspect ratio. Additionally, they noted that while neighboring pore-throat correlations significantly impacted the shape of k_r curves, the overall pore-size distribution had only a minor influence. More recently, Mukherjee et al. (2022) examined a similar reversal behavior in k_r trends and linked it to interfacial area dynamics. This study highlights the importance of the relationship between the corners and areas of the throat in a network model. They observed that during imbibition, a significant portion of the total interfacial area was associated with the corner regions, whereas during primary

drainage, the interfacial area was predominantly concentrated in the pore throats. This shift in the distribution of the interfacial area was proposed as a key factor that influences k_r reversal. Our experimental results substantiate this claim by demonstrating that the reverse k_r behavior occurs in agreement with the reverse behavior in the A_{nw} hysteresis.

The k_r reversal phenomenon has also been reported in subsurface contamination studies involving dense non-aqueous phase liquids (DNAPLs). As noted by Grant et al. (2005), contaminated groundwater plumes from persistent source zones often exhibit non-wetting phase k_r curves where imbibition permeabilities exceed those in drainage. These observations further suggest that the underlying cause of the reversal is not universal to gas-brine systems only. Taken together, our study and the aforementioned works indicate that the reversal behavior in k_r is not only a function of the consolidation state of the medium but is also closely related to the pore structure and the phases used, that is, oil/water or gas/water. These factors collectively govern how fluids move during drainage and imbibition, leading to the observed hysteretic behavior.

5. Conclusions

In this study, we performed a comparative analysis of k_r hysteresis in oil-brine and gas-brine systems using synchrotron-based micro-CT imaging. Our results indicate that for the gas-brine system, there is a reversal behavior in the placement of drainage and imbibition phases compared to oil-brine systems. Specifically, gas-brine imbibition tends to occur above drainage in the interfacial area-saturation relationship, suggesting that gas trapping and recovery processes behave differently than in the similarly tested oil-brine system. To investigate the factors that influence hysteresis in each system, we examined pore-scale characteristics, including S_w , P_c , A_{nw} , and χ_n . Our findings demonstrate that k_r hysteresis is strongly affected by A_{nw} . In particular, trends in A_{nw} show significant differences between the tested oil-brine and gas-brine systems, highlighting the need for a new modeling approach to capture the observed reverse hysteresis. Therefore, we propose a new hysteresis model that can capture the reversal behavior of our gas-brine system. The factors influencing the reserve behavior were discussed in detail and compared with those in other published works, highlighting that such behavior is caused by a combination of the phases used and the pore structure, with such behavior expected for unconsolidated gas-brine systems. Although the applicability of our proposed hysteresis model depends on the specific application and would need to be verified on a case-by-case basis, it is clear that such behavior is linked to how much specific interfacial area is generated during imbibition versus drainage.

The findings of this work will have implications for multiphase flow in porous media, which plays a key role in various scientific and engineering applications. In particular, understanding the hysteresis in k_r can provide valuable insights into improving the selection and design of porous materials in systems where efficient gas transport is crucial. By adjusting the pore structures to minimize hysteresis effects or reverse behavior, porous media can be designed to enhance gas mobility and ensure more predictable flow. While this work offers valuable insights at the pore-scale, industrial applications will need to account for factors such as medium heterogeneity, scalability, and the efficiency of gas transport under varying conditions. As a potential direction for future work, investigating the impact of different rock types on hysteresis behavior could refine our understanding of the reversed k_r behavior observed in gas-brine systems. Additionally, conducting similar experiments using different gases such as CO_2 , CH_4 , and N_2 could help determine whether the reversed relative permeability behavior observed with air persists across various gas types and rock-fluid interactions. Future studies could explore how variations in rock texture, porosity, and permeability influence hysteresis and gas transport behavior, offering deeper insights into the distinct flow characteristics of gas-brine and oil-brine systems and improving our understanding of their k_r behavior.

CRediT authorship contribution statement

Ahmed Adila: Writing – original draft, Visualization, Validation, Software, Formal analysis. **Mohammad Ebadi:** Writing – original draft, Supervision, Investigation. **Wen Xi:** Methodology, Data curation. **Xiang Qi:** Methodology, Data curation. **Yu Jing:** Formal analysis, Data curation, Conceptualization. **Peyman Mostaghimi:** Validation, Supervision, Formal analysis, Conceptualization. **Ryan T. Armstrong:** Writing – original draft, Validation, Supervision, Investigation, Formal analysis, Conceptualization.

Open research

The oil-brine system images of Schlüter et al. (2016) are available at <https://www.ufz.de/record/dmp/archive/5732/de/>

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.ijmultiphaseflow.2025.105483>.

Data availability

Data will be made available on request.

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